INCH-POUND
NQ/PD 26-01
10 April 2026
DRAFT

PURCHASE DESCRIPTION
BODY ARMOR, IMPROVED MARITIME BALLISTIC PLATE (IMBP)

This purchase description (PD) is approved for use by all Departments and Agencies of the Department of Defense (DOD). Recommended improvements, simplifications, or reductions in paperwork are encouraged and should be directed to the preparing activity.

1. SCOPE

1.1 Scope. This purchase description covers the requirements for an Improved Maritime Buoyant Plate (IMBP) to provide protection against rifle threats for U.S. Navy forces while performing Visit, Board, Search and Seizure (VBSS) and Anti-Terrorism (AT) duties. When placed in the Maritime Armor System (MAS), Naval Security Forces (NSF) Tactical Vest (NTV) or other compatible armor systems, the IMBP will provide protection from specific 5.56mm and 7.62mm rifle rounds. The IMBP is a critical safety item.

1.2 Classification. IMBP shall be one type in the following sizes:

X-Small (XS), Small-Short (SS), Small (SM), Medium (MD), Large (LG), and X-Large (XL).

Controlled by: Naval Sea Systems Command
Controlled by: PMS 408 Expeditionary Missions
CUI Category: CTI
Distribution/Dissemination Control: Distribution D
POC: PMS 408 Expeditionary Missions

Comments, suggestions, or questions on this document should be addressed to Program Manager Expeditionary Missions, 1333 Isaac Hull Ave, Washington Navy Yard, DC 20376. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this purchase description. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements, whether or not they are listed.

2.2 Government Documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

CO/PD 04-19	-	Personal Armor, Enhanced Small Arms Protective Insert (ESAPI)
MIL-DTL-32075	-	Label: For Clothing, Equipage, and Tentage (General Use)
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-662	-	V_{50} Ballistic Test for Armor
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-3027	-	Performance Requirements and Testing of Body Armor
NQ/PD 16-03	-	Body Armor, Maritime Armor System (MAS)
NQ/PD 23-01	-	Body Armor, Naval Security Forces (NSF) Tactical Vest (NTV)

(Copies of these documents are available upon request).

2.2.2 Other Government Documents, Drawings, and Publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

Drawing Numbers:
2-6-0588 REV A7; Enhanced Small Arms Protective Insert, Size; X-small
2-6-0589 REV A7; Enhanced Small Arms Protective Insert, Size; Small
2-6-0594 REV A2; Enhanced Small Arms Protective Insert, Size Small Short
2-6-0590 REV A7; Enhanced Small Arms Protective Insert, Size; Medium
2-6-0591 REV A7; Enhanced Small Arms Protective Insert, Size; Large
2-6-0592 REV A7; Enhanced Small Arms Protective Insert, Size; X-large

TEST OPERATING PROCEDURES

TOP 10-2-210 Ballistic Testing of Hard Body Armor Using Clay Backing

(Requests for USATECOM TOP 10-2-210 shall be made to Commander, US Army Aberdeen Test Center, ATTN: TEDT-AT-SL-T, 400 Colleran Road, Aberdeen Proving Ground, MD 21105-5059)

GUIDES

Security Classification Guide (SCG) for Soldier Protection System (SPS) & Associated Soldier Protective Equipment (SPE) V19.0 dated 03 Sep 2019

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

DIRECTOR, OPERATIONAL TEST & EVALUATION (DOT&E)

4-27-10 Standardization of Hard Body Armor Testing
7-2-10 Standard for Lot Acceptance Ballistic Testing of Hard Body Armor

2.3 Non-Government Publications. The following documents forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC Method 169 - Weather Resistance of Textiles

(Applications for copies should be addressed to the American Association of Textile Chemists and Colorists, PO Box 12215, Research Triangle Park, NC 27709-2215 or www.aatcc.org)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D792 -	Specific Gravity and Density of Plastics by Displacement
ASTM D1141 -	Standard Practice for the Preparation of Substitute Ocean Water
ASTM D2563 -	Standard Recommended practice for Classifying Visual Defects in Glass Reinforced Plastic Laminate Parts
ASTM D2584 -	Ignition Loss of Cured Reinforced Resins
ASTM D2863 -	Measuring the Minimum Oxygen Concentration to Support Candle Like Combustion of Plastics (Oxygen Index)
ASTM D3951 -	Standard Practice for Commercial Packaging
ASTM E29 -	Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications

(Copies of these documents are available online at www.astm.org or from ASTM International, 100 Bar Harbor Drive, West Conshohocken, PA 19428-2959.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQ Z1.4-2008 -	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org or from the American Society for Quality, P.O. Box 3005, Milwaukee, WI, 53201-3005.)

NATIONAL INSTITUTE OF JUSTICE (NIJ)

NIJ 0101.07 -	Ballistic Resistance of Body Armor

(Copies of this document are available online at http://www.justnet.org/ or from NLECTC4 National, 2277 Research Blvd., M/S 8J, Rockville, MD 20850.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence over TOP and TOP takes precedence over MIL-STD-662. However, for V50 testing, MIL-STD-662 takes precedence over TOP. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. First Article Testing (FAT) is a requirement for IMBP. Samples shall be subjected to FAT in accordance with paragraph 4.1.1.

3.2 Construction. The materials selected by the contractor shall be capable of meeting all the performance, operational and environmental requirements specified herein.

3.2.1 IMBP Cover. The IMBP cover materials shall mitigate frontal spall, increase durability, and decrease environmental deterioration of the underlying components configuration. The cover material shall cover the entire front and back outer surface and all side edges. The cover, including all edges, shall be free of cuts, holes, tears, abraded areas, wrinkles, bubbles or creases. The cover shall not exhibit any areas of under adherence. There shall be no visible peeling, wrinkles, bubbles or creases of the cover before or after any operating environment conditioning when verified in accordance with section 4.5.

3.2.1.1 Covering of Components by Spray Coating. The IMBP shall exhibit uniform coverage of any spray coated material. The finished plate shall exhibit no smears and/or clumps on the exposed surfaces or spray coated material, nor exhibit any under covered areas on exposed surfaces.

3.2.2 Assembly of Components by Bonding. The IMBP shall exhibit no evidence of under adherence between layers or components. The finished IMBP shall exhibit no smears and or clumps from excess bonding agent visible on the exposed surfaces when verified in accordance with 4.2.

3.3 Production Data. The following information, determined during production, shall be made a matter of record and shall be furnished on request to the contracting official. This data shall be identified with the serial number of the body armor.

a. Data generated during inspection or other protocols per quality system/quality validation plan. This includes, but is not limited to, weight, thickness and dimensional measurements.

b. Supplier lot information and traceability for all component parts identified in the technical data package. This shall include material compliance forms signed by the contractor, each sub-contractor or material supplier.

c. Operational, ownership and environmental test data generated by the contractor on the IMBP.

d. Ballistic performance test data generated under all first article, conformance and validation testing as described in paragraph 4.8.

e. For traceability, every IMBP shall be durably marked in such a fashion as to be traceable from production through to the ballistic test records for that lot of IMBPs. The serial number shall be marked on the outside face within 1 inch of the lower edge of any ceramic component (if used) before it is fired. It shall be marked by inscribing, embossing, or with high temperature resistant inorganic ink after firing with permanent marking to a maximum depth of 0.010 inch so that the serial number will be legible when the cover is stripped mechanically or by the use of a solvent. Solvents, fuels and other liquids shall not affect the serial number markings. It is desirable to have ceramic cores marked to be traceable by radiographic analysis as well. If inscribed, it will be done on the outer 0.25 inches of the plate.

3.4 Workmanship. The finished IMBP shall conform to the quality of product established by this purchase description. Utmost care shall be taken during fabrication to ensure quality workmanship and safety of the service person using the item. All materials to be used in the construction of the IMBP shall conform to all material requirements in this purchase description and internal specifications, unless otherwise specified. Manufacturing practices shall be capable of consistently yielding product that conforms to all requirements in this purchase description and internal specifications for the body armor and their components. Continual improvement shall also be a constant focus of the manufacturing practice. All component materials shall be properly marked and identified and shall be protected properly during storage. Materials shall be produced and integrated to extend durability and provide consistency of appearance throughout plate life. All components to be assembled shall be thoroughly cleaned of all foreign matter. Surfaces to be bonded shall be properly prepared in a manner which will ensure a proper bond capable of meeting the applicable performance requirements. The required adhesive(s) shall be applied uniformly over the entire contact areas of the components to be joined to eliminate delamination between materials. Material layers shall be free of contaminants (such as, but not limited to: Foreign Object Debris (FOD) (media not associated with the technical data package), loose fragments of component materials, operator elements not part of component materials. Any IMBP ceramic (if used) must be free of cracks. Defective conditions within the IMBP shall not exceed the limits identified in Table D.1 (Appendix D). This section is applicable to all material or components of the IMBP whether furnished by the Prime Contractor or by any of their suppliers or sub-contractors. In any case of material, process, or equipment change desired to be made by the Contractor, the written consent of the Government shall be obtained before making the change. Additional testing may be required prior to implementation of the change to verify product performance. (See section 4 for verification).

3.4.1 Uniform Areal Density. The IMBP shall provide uniform areal density (materials coverage) throughout the entire surface area. The only exception shall be when edge materials are used to protect the plate from side impact (e.g., foam) to increase the durability. Material no thicker than 0.10 inches (2.54 mm) may be used for this purpose. Materials shall extend from edge to edge to provide uniform thickness throughout the entire surface area of the IMBP. “Patches,” “clamps,” and other materials with partial or interrupted coverage of the IMBP surface area shall not be acceptable. Materials shall not cause interference with Non-Destructive Test Equipment (NDTE) (e.g., radiographic inspection or computed tomography). If ceramic materials are used, the ceramic shall have uniform edges. To reduce sharp edges, an abrasive device or other equipment may be used to clean the edges in production. Uniform areal density shall be verified in accordance with paragraph 4.2.2.3.

3.5 Operational Requirements. IMBP are neutrally buoyant, multi-curve, hard armor plates intended for use in multiple armor systems. Each IMBP shall meet the following user-oriented requirements (see paragraph 4.5).

3.5.1 Ease of Insertion. The IMBP shall be able to easily slide into and out of the corresponding sized Maritime Armor System (MAS) and Naval Security Forces (NSF) Tactical Vest (NTV) plate pockets. No special training, equipment or tools will be required to insert the IMBP into the plate pocket (see paragraph 4.5.1).

3.5.2 Weight. The finished IMBP threshold (maximum) and objective (desired) weights are outlined in Table 1 (rounded to the nearest 0.1 lb) (see paragraph 4.5.2).

Table 1. IMBP Maximum Weight

Size	Threshold Weight (lbs)	Objective Weight (lbs)
X-Small	3.00	2.40
Small-Short	3.10	2.80
Small	3.60	3.00
Medium	4.20	3.40
Large	4.80	4.00
X-Large	5.40	4.60

3.5.3 Dimensional Measurements. The IMBP shall conform to all measurements, tolerances, radius and edge chamfers as specified in drawings listed in paragraph 2.2.2 (see paragraph 4.5.3). Appendix D outlines applicable out of tolerance grading scale for determination of acceptance.

3.5.4 Thickness. The IMBP in finished form shall have a maximum 1.0-inch uniform thickness throughout. The variability of the local thickness on any part shall not exceed 1/8 inch (see para. 4.5.4).

3.5.5 IMBP Color. All outer areas of the IMBP shall be colored Black and be a good match to the standard sample (see paragraph 4.5.5).

3.5.6 Labels/Markings. All labels and markings shall be approved by NAVSEA prior to production. (See paragraph 4.5.6).

3.5.6.1 Strike Face Labels. The size identification of the IMBP shall be clearly displayed centered on the front surface 1.5 inches below the top edge in all capital characters 0.5 inch high (i.e. MEDIUM). A permanent label or marking shall be centered on the front/strike face, 3.0 ± 0.25 inches from the top clearly displaying “STRIKE FACE”. All characters will be 0.50-inch height. The words “HANDLE WITH CARE” shall be printed in characters 0.5-inch-high at 6.5 ± 0.25 inches below the top edge and centered on the front face surface.

3.5.6.2 Unique Identification and Instruction Label. As a threshold the IMBP shall have a combination unique identification (UID) and markings or label with the following detailed requirements:

a. The Human Readable Information (HRI) contents of the UID label shall be as follows:
   1. Name of Ballistic Plate
   2. Size
   3. National Stock Number (NSN) / Contract Number / Lot Number / Serial Number / Date of Manufacture / Manufacturer Commercial and Government Entity (CAGE) code / Design Code

b. Machine Readable Information (MRI) contents of the UID label shall consist of one ECC 200 compliant Data Matrix code containing: CAGE code of manufacturer, Lot Number, Serial Number, Date of Manufacture, NSN, and Design Code. The tag/label shall comply with the following:

   1. Data Matrix Construct: The Data Matrix shall be encoded per MIL-STD-130 using only the data identifiers (DI) and criteria shown below. The following DI sequence shall be maintained in the order listed below:
      - Cage=17V followed by cage code
      - Lot=1T followed by lot number
      - Serial number=S followed by serial number
      - Date of production = 16D followed by production date, YYYYMMDD
      - National stock number=N followed by the NSN.
      - Part number = 1P followed by design code (the design code

may be up to 13alpha-numeric characters (plus only dashes "-" as special characters))

- Construct Example:
- [)>RS06GS17V52969GS1TE034GSS328185GS16D20080215GSN8470-01-520-7370GS1PABC-123RSEOT

2. Data Matrix Geometry: Data Matrix codes shall be a square ECC200 matrix per ISO 16022. Individual Cell size (element size) of the code shall be between 0.020 and 0.023 inches. A quiet zone of 0.5 inches of Black label/tag material is required around the Data Matrix code.

c. The instruction markings or label shall include the appropriate designation for each plate as to placement of the plate in relation to the body when worn. Each marking or label shall be centered on the rear of the ballistic plate so it may be easily read when the plate is extracted from a plate pocket. The contents of the instruction label shall include the following verbiage:

1."THIS SIDE AGAINST BODY" on the rear of the ballistic plate.
2."Remove loose dirt from surface with a cloth or soft brush."
3."The Lightweight Plate is unserviceable if any of the following conditions are present:

- Outer cover is damaged exposing inner materials.
- Plate is cracked (e.g., audible rattling when shaken).
- Cracking is felt or heard when 0.50 in (12.7 mm) perimeter of plate is firmly pinched.
- Creaking or squeaking is heard when twisted by hand.
- Composite is delaminating (backing materials are separating).
- Plate has been hit by a bullet or fragment."

d. The color of the label shall be black with white lettering/marking. All plate label and spall colors shall be approved by NAVSEA prior to production.
e. The plate markings or label shall remain durable, legible, and affixed to the rear of the ballistic insert during field use under all environmental conditions. The label shall be of sufficient strength to withstand repeated abrasion during field use and cleaning when verified in accordance with the weatherometer conditioning in paragraph 4.7.1.
f. The plates shall be delivered with unique item identifiers in accordance with the current version of DFARS clause 252.211-7003., "Item Unique Identification and Valuation", ANSI MH10.8.2-2013, "Data Identifier and Application Identifier Standard" and MIL-STD-

130, “Identification and Marking of U.S. Military Property.”

1. AS9132 and AIM DPM grading platforms will not be allowed for this project. Contractor must provide the contracting officer with at least two verification reports per plate size for each FAT and LAT. Verification shall be on the end item, not a standalone label. If using laminates or overcoats the label must be verified after placing the laminate or overcoat on the label or tag. No exceptions are allowed. Proof of Verification is subject to inspection at the time of shipment.

2. Validation: Validation checks of the UID must be performed on a routine basis. Contractor is responsible for encoding the UID per above guidelines and the latest revision of MIL-STD-130. Proof of Validation is subject to inspection at the time of shipment.

3. Placement of the UID label/tag: The center of the Data Matrix code on an x and y axis will hereinafter be referred to as the centerline of the UID label/Tag. The UID label/tag will be placed on the back side (opposite the strike face) of the item. The centerline of the UID label/tag will be left-right centered on the item, positioned 2 inches from the bottom edge. A tolerance of ± 0.25 inches in each direction will be allowed. The entire plate area within 3.0 inches of the bottom of the plate must be clear of any other labels or markings at all times. Additional non UID information and logos required by this product description must appear above the 3-inch featureless zone.

3.5.7 Health and Safety. The IMBP shall be non-hazardous (non-explosive and have no toxicological or electromagnetic radiation effects) to the individual wearing the IMBP.

3.5.7.1 Personnel Hazard. The vendor shall visually inspect that IMBPs are free of sharp edges, foreign materials, burrs or any other conditions that may cause an injury to the user. The vendor shall demonstrate that the IMBP will not cause injury to the user, or surrounding sailors, when ballistically tested (paragraph 4.8).

3.5.7.2 Magnetic Influence. There shall be no magnetic influence on a compass at any distance from the IMBP (paragraph 4.5.7.2).

3.6 Ownership and Support. (See paragraph 4.6)

3.6.1 Durability. The IMBP shall be able to withstand two drops using a moment arm fixture at a height of 48.0-inches onto a concrete surface without any detrimental effects to ballistic performance, major surface characteristics or physical properties. A 10.0 ± 0.1 lbs weighted object shall be attached to the rear surface of the IMBP sample (see paragraph 4.6.1). The weight of the moment arm and straps shall be 8.5 ± 0.5 lbs.

3.7 Operating Environment Requirements. Unless otherwise stated, ballistic validation testing will be performed against threats outlined in Appendix A and Appendix B. Ballistic performance shall not be degraded from exposure to the environmental conditions specified below.

3.7.1 Weatherometer Resistance. The finished IMBP after being subjected to the weatherometer resistance testing shall exhibit no evidence of cracking, blistering, color change, separation of edging (if used) or ballistic degradation when conditioned in accordance with paragraph 4.7.1.

3.7.2 Temperature Extreme. The IMBP shall provide structural and ballistic protection when exposed to temperatures ranging from -60 ºF to +160 ºF when conditioned in accordance with paragraph 4.7.2.

3.7.3 Fluid Resistance. The IMBP shall maintain structural and ballistic integrity after contamination with Jet Propellant 8 (JP-8), oil, and immersion in salt water for a period of two (2) hours each at 70 degrees ± 10 degrees Fahrenheit (see paragraph 4.7.3).

3.7.4 Altitude. The IMBP shall meet all performance requirements from sea level to 15,000 feet (ft.) altitude. No structural, visible, operational degradation, or safety hazard shall occur when the IMBP is exposed to a pressure change equivalent to a change in altitude from 40,000 ft. to 15,000 ft. (12192 meters (m) to 4572 m) at a rate of 1500 ft./min - 2000 ft./min (457.2 m/min - 609.6 m/min) (see paragraph 4.7.4).

3.7.5 Temperature Shock. The IMBP shall meet all performance requirements after exposure to temperature changes between the high and low operating temperature extremes; temperature shock shall occur within a five minute period when conditioned in accordance with paragraph 4.7.5.

3.7.6 Buoyancy. The IMBP shall have a buoyancy of greater than or equal to zero when tested per 4.7.6 and Appendix E.

3.8 Performance Requirements. (See paragraph 4.8)

3.8.1 Area of Coverage. The IMBP will provide uniform materials coverage throughout the entire surface area of the IMBP. If backing materials are used, the backing materials must extend from edge to edge to provide uniform thickness throughout the entire surface area. The IMBP shall have uniform areal density throughout the entire surface area. “Patches”, “clamps”, materials with partial coverage of the IMBP surface area shall not be acceptable. Any cuts with open gaps and/or slits on any materials are not allowed. The government has the option to reject plates with imperfections visible in radiographic images (x-rays). If ceramic materials are used, the ceramic shall have uniform edges without “edge breaks” or “chamfers”. To reduce sharp edges, sandpaper or a scouring pad may be used to clean the edges in production.

3.8.2 Required V50 Ballistic Limit. The IMBP will yield the minimum V50 ballistic limit measurements at 0-degree obliquity when tested per paragraph 4.8.6 with the specified test projectiles as per Appendix A and Appendix B.

3.8.3 V0 Ballistic Resistance. The IMBP will provide ballistic resistance as specified in Appendix B when tested per paragraph 4.8.7.

3.8.4 Back Face Deformation. The IMBP back face deformation shall be tested in accordance with and meet the requirements of paragraph 4.8.9 and Appendix B.

3.8.5 Shatter Gap Resistance. The IMBP will provide ballistic resistance against shatter gap per paragraph 4.8.10 and Appendix B.

4 VERIFICATION

4.1 Classification of Testing. The testing requirements specified herein are categorized as First Article Testing (FAT) and Lot Acceptance Testing (LAT) as specified in Table 2.

Table 2. Requirements and Verifications

CHARACTERISTICS	REQUIREMENT PARAGRAPH	VERIFICATION PARAGRAPH	First Article Test (FAT)	Lot Acceptance Testing (LAT)
Construction Requirements	**3.2**	**4.2**		
IMBP Cover	3.2.1	4.2.2	X	X
Assembly of Components by Bonding	3.2.2	4.2.2	X	X**
Workmanship Requirements	**3.4**	**4.4**		
Areal Density	3.4.1	4.4	X	X**
Operational Requirements	**3.5**	**4.5**		
Ease of Insertion	3.5.1	4.5.1	X	X**
Weight	3.5.2	4.5.2	X	X

Dimensional measurements	3.5.3	4.5.3	X	X
Thickness	3.5.4	4.5.4	X	X
IMBP Color	3.5.5	4.5.5	X	X**
Labels/markings	3.5.6	4.5.6	X	X**
Health and Safety	3.5.7	4.5.7		
Personnel Hazard	3.5.7.1	4.5.7.1	X	CoC
Magnetic Influence	3.5.7.2	4.5.7.2	X	X**
Ownership and Support Requirements	**3.6**	**4.6**		
Durability	3.6.1	4.6.1	X	NA
Operating Environment	**3.7**	**4.7**		
Weatherometer	3.7.1	4.7.1	X	NA
Temperature Extreme	3.7.2	4.7.2	X	NA
Fluid Resistance	3.7.3	4.7.3	X	NA
Altitude Test	3.7.4	4.7.4	X	NA
Temperature Shock	3.7.5	4.7.5	X	NA
Buoyancy	3.7.6	4.7.6	X	X
Performance Requirements	**3.8**	**4.8**		
Area of coverage	3.8.1	4.8	X	NA
Req V_{50} Ballistic Limit	3.8.2	4.8.6	X	X***
V_0 Ballistic Resistance	3.8.3	4.8.7	X	X
Back Face Deformation	3.8.4	4.8.9	X	X
Shatter Gap Resistance	3.8.5	4.8.10	X	X***

Notes:

X - Testing required
CoC - Certificate of Conformance to Include data
** - Test or inspection to be performed at vendor facility
*** - The government may elect to perform testing at their discretion
NA - Not Applicable

4.1.1 First Article Testing (FAT). The government reserves the right to perform any of the tests set forth in this specification where such tests are deemed necessary to ensure supplies and services conform to prescribed requirements. When a FAT is required, it includes all of the verifications listed in Table 2 unless otherwise specified in the contract.

4.1.2 Lot Acceptance Testing (LAT). LAT sample test size shall be determined using ANSI/ASQ Z1.4-2008, Special Inspection Level S-4, AQL of 4, single sampling plans for normal inspection. LAT of IMBP shall include those applicable examinations and tests outlined in Table 2. Refer to Appendix C for LAT Protocol.

4.2 Requirements and Verifications. Table 2 delineates performance requirements and the associated verification methods.

4.2.1 Order of Testing. Performing the various testing (operating, ownership & support, operating environment, and performance) can occur in any order.

4.2.2 Verification Methods. The types of verification methods included in this section are visual inspection, dimensional measurements, sample tests, component properties analysis and similarity to previously approved or previously qualified designs.

4.2.2.2 Verification Using Standard Samples. Use standard samples to verify colors with visual inspections.

4.2.2.3 Uniform Areal Density. Dry lay-up for each design shall be examined to visually ensure the plates exhibit uniform areal density (no overlap or underlay, uniform material lengths, and thicknesses, etc.).

4.3 Responsibility for Compliance. Production items shall meet all requirements specified in section 3. The supplier shall establish and maintain documented procedures for inspection and testing activities in order to verify that the specified requirements for the product are met. The required inspection, testing and the records to be established shall be detailed in a quality plan available to the government as specified in the contract or procuring activity. The inspection set forth in this specification shall become part of the contractor's overall inspection procedures or quality system. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection as part of the manufacturing operations is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.4 Workmanship and Areal Density. Dry Lay ups for each design will be examined prior to production of any test plates or production plates to visually ensure the plates exhibit uniform areal density (no unnecessary overlap/underlay, uniform material lengths and thicknesses, etc.). Radiographic images (x-rays) of complete end items will be taken digitally or by the use of x-ray film. Radiography shall include the entire strike face area of the plate with the radiograph taken from a frontal or rear view. Defects will be classified in per Table D.1 (Appendix D) and grading criteria for acceptance of defects is outlined in Appendix D. The Government reserves the right to change the shot pattern to target any and all anomalies, defective conditions or contaminants highlighted in x-ray. If a failure is observed when tested in accordance with sections 4.6, 4.7, and 4.8 the items shall be rejected regardless of defect classification specified in Table 2. Visual end item inspection shall take place to ensure the entire plate is covered completely with no rips, tears, cuts, blistering or separation between the materials.

4.5 Operational Requirements. Complete each test in this paragraph.

4.5.1 Insertion. One barehanded person shall demonstrate insertion of the appropriately sized IMBP into the Naval Security Forces (NSF) Tactical Vest (NTV) plate pockets without tools or special aids. Perform insertion into Government furnished carrier in a maximum of 30 seconds.

4.5.2 Weight. Take physical weight measurements of all IMBPs to ensure that weights do not exceed those presented in paragraph 3.5.2 Table 1. The finished IMBPs shall be weighed to the nearest 0.1 pound.

4.5.3 End Item Dimensions. Take physical measurements of all IMBP to show that all measurements conform to dimensions and tolerances in Drawings specified in paragraph 2.2.2. Appendix D outlines applicable out of tolerance grading scale for determination of acceptance

4.6.3.1 Physical measurements of all submitted IMBPs will be obtained using an ATOS Scanbox with a FANUC Robot M-20iA and an ATOS Triple Scan Sensor to show that all measurements conform to the dimensions and tolerances per drawings outlined in paragraph 2.2.2. In the event of system failure, a laser scanning device (e.g., FARO arm) may be used in lieu of the ATOS Scanbox to measure the physical dimensions of the submitted IMBPs.

4.5.4 Thickness. All IMBPs will be measured for thickness using an ATOS Scanbox with a FANUC Robot M-20iA and an ATOS Triple Scan Sensor. All measurements will be recorded to the nearest 0.01 inch. Thickness will be measured near the center (at least 3.5 inches from any edge) and near each of the six corners (1/4 to 1 inch from the edge) of each IMBP. If required, precision calipers may be used in lieu of the ATOS Scanbox to measure thickness of the submitted IMBPs.

4.5.5 IMBP Color. Visually inspect the IMBPs for similarity to the standard sample color of black as specified by the contract.

4.5.6 Labels.

4.5.6.1 Visually inspect for legible, permanent labels on the back face surfaces. Visually inspect that labels contain Human Readable Information (HRI) listed in paragraph 3.5.6. The durability of label markings shall be checked with the following procedures:

a) A representative area of the label markings shall be rubbed by hand for 15 seconds with a cotton cloth soaked in distilled water.
b) The same area shall then be rubbed by hand for 15 seconds with a cotton cloth soaked with denatured alcohol (methylated spirit).
c) Finally, the same area shall then be rubbed by hand for 15 seconds with a cotton cloth soaked in isopropyl alcohol.

4.5.6.2 Using appropriate IUID data matrix reader, verify UID construct data elements are per 3.5.6.2.

4.5.7 Health and safety. Complete the verifications in this paragraph.

4.5.7.1 Personnel Hazard. Visually inspect that the items are free of conditions that will cause potential injury to the user. All the material comprising the end item shall be examined by the Government to assess non-explosive, toxicological, and electromagnetic radiation effects. The Contractor shall provide Material Safety Data Sheets (MSDS) to the Government for all materials comprising the end item.

4.5.7.2 Magnetic Influence. The IMBP shall be tested for magnetic influence using magnetic and lensatic compasses. The IMBP shall not cause any deviation of the compass needle.

4.6 Ownership and Support Requirements. Perform the verification in this paragraph.

4.6.1 Durability. Demonstrate durability by performing impact test as outlined in TOP 10-2-210.

4.7 Operating Environment Requirements. Perform each verification in this paragraph then perform ballistic testing in accordance with Appendix B. All test items shall be pre-conditioned for 24 hours at 70ºF±10ºF at relative humidity of 30%±10% prior to being subjected to subsequent conditioning.

4.7.1 Weatherometer Resistance. The IMBP shall be tested for weather resistance as outlined in TOP 10-2-210.

4.7.2 Temperature Extreme. IMBP will be tested after exposure to temperature per TOP 10-2-210.

4.7.3 Fluid Resistance. IMBP will be tested after fluid exposure to Salt Water, Lubricating oil and F-24 as outlined in TOP 10-2-210.

4.7.4 Altitude Test. IMBP will be tested after simulated altitude exposure per

TOP 10-2-210.

4.7.5 Temperature Shock. IMBP will be tested after temperature cycling as outlined in TOP 10-2-210.

4.7.6 Buoyancy Test. IMBP buoyancy will be measured per Appendix E.

4.8 Performance Requirements. Complete each test in this paragraph.

4.8.1 Ballistic Test Criteria. For all V50 BL and V0 acceptance tests the information outlined in TOP 10-2-210, Section 5, is required by the government to validate performance

4.8.2 Projectile Velocity Determination. Projectile velocity determination will be per TOP 10-2-210.

4.8.3 Weapon Mounting Configuration. Weapon mounting configuration will be per TOP 10-2-210.

4.8.4 Environmental Test Conditions. Environmental test conditions will be per TOP 10-2-210.

4.8.5 Projectile Yaw. Projectile yaw determination will be per TOP 10-2-210.

4.8.6 V50 BL. See Appendix A and Appendix B.

4.8.6.1 PP and CP for V50. Complete and partial penetrations will be determined based on the definitions provided in paragraph 6.3.

4.8.6.2 V50 Test Sample Mounting. Unless otherwise stated the following conditions shall be performed during V50 testing. The IMBP shall be secured to the clay-mounting block with the impact side perpendicular to the line-of-flight of the projectile. Testing will be performed in accordance with MIL-STD-662 except partial and complete penetrations will be determined based on definitions provided in paragraph 6.3.

4.8.7 Ballistic V0 Testing. Ballistic V0 testing of IMBP shall be conducted on a recurring basis per paragraph 3.8.3 and Appendix C.

4.8.7.1 V0 Determination for Acceptance. For V0, the minimum velocities as stated in Appendix B will be the requirement.

4.8.8 Impact Location. See Appendix A.

4.8.9 Back Face Deformation (BFD) Measurement. BFD Measurements will be per TOP 10-2-210.

4.8.9.1 BFD Measurement Process. The clay material shall be contoured to the back face curvature provided by the Lightweight Plate before ballistic testing. This buildup shall use additional clay backing material conditioned in the same manner as the clay material fixture. Both the shootpack and Lightweight Plate shall be strapped or taped to the curved surface of the clay material. BFD shall be measured from the original undisturbed surface of the clay backing material to the post-impact surface with the reference direction perpendicular to the front surface (facing the line-of-fire) of the box. BFD measurements shall be conducted for all shots. Indentation measurements shall utilize laser scanner measurement instruments, which provide a means to accurately establish the difference between the original undisturbed clay surface, and the post-impact surface. The BFD measurement is the maximum-distance-length, which is the length of the longest line segment parallel to the reference direction between the pre-impact clay surface and the post-impact (BFD) clay surface, where the reference direction is defined to be perpendicular to the front surface (facing the line-of-fire) of the box containing the clay backing material. Deformations shall be recorded in mm to the nearest tenth digit following standard ASTM E29 “Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications” (“Five-even” rule)(Rounding Method) (i.e. 48.050 = 48.0, 44.051 = 44.1 and 47.950 = 48.0)

4.8.10 Shatter Gap Testing. Per Appendix B, mounting per TOP 10-2-210.

5 PACKAGING

5.1 Packing. For acquisition purposes, the contract or order shall specify complete packaging requirements. When DOD personnel perform material packaging, those personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. The Inventory Control Point packaging activity within the Military Department of defense Agency, or within the Military Department’s Systems Command, maintains packaging requirements. Packaging data retrieval is available from the managing Military Department’s or defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended Use. The IMBP is intended for use in the Naval Security Forces (NSF) Tactical Vest (NTV) and the Maritime Armor System (MAS) by Force Protection (FP) personnel and Visit Board Search and Seizure (VBSS) teams. The IMBP will provide multi hit protection against specified small arms.

6.2 Acquisition Requirements. Acquisition documents should specify the following:

a) Title and date of this document.
b) When first article, and pre-production items are required.
c) Color required.
d) Size tariff.
e) Contractually approved ballistic package(s) to include package name, complete description, and FAT acceptance letter.

6.3 Ballistic Testing Definitions. MIL-STD-3027 shall be used for definition references only. In the case of a conflict between MIL-STD-3027 and this document, this purchase description shall take precedence. The following definitions are provided to assist in understanding the test procedures:

Over-Velocity. Striking velocity that is higher than the specified threat requirement.

Under-Velocity. Striking velocity that is lower than the specified threat requirement.

Complete Penetration (CP) for Acceptance Testing. A complete penetration will have occurred when the projectile, fragment of the projectile or fragment of the armor material is imbedded or passes into the clay backing material used to measure transient deformation. Paint or fibrous materials that are emitted from the back of the test specimen and rests on the outer surface of the clay impression are not considered a complete penetration.

Fair Impact. A projectile that impacts the armor within the specified velocity and yaw shot spacing requirements, will be considered a fair impact.

Partial Penetration PP(P). Any fair impact that is not a complete penetration shall be considered a partial penetration.

Crown. Location of the intersection of three different curvatures vicinity the center of the upper third of the plate.

Areal Density (AD). A measure of the weight of the ballistic filler of the armor panel per unit area, usually expressed in pounds per square foot (lb/ft^2) or kilograms per square meter (kg/m^2)

of surface area.

Obliquity. A measure, normally in degrees, of the extent to which the impact of a projectile on an armor material deviates from a line normal to the target. Thus, a projectile fired perpendicular to an armor surface at 0 degrees obliquity.

Spall. Fragmentation of the bullet or target material which is projected from the impact surface or rear surface of the target.

Yaw. Projectile yaw is the angular deviation of the longitudinal axis of the projectile from the line of flight at a point as close to the impact point on the target as is practical to measure.

V50 Ballistic Limit (BL). In general, the velocity at which the probability of penetration of an armor material is 50 percent.

6.4 Government Loaned Property. Contact the contracting official if necessary for the loan of property required to verify conformance with paragraph 3.5.1.

6.5 Drawings and Materials. See Paragraph 2.2.2.

6.6 National Stock Numbers. The following are the National Stock Number (NSN) or Navy Item Control Number (NICN) assigned for IMBP.

Table 3. IMBP NSN Data

IMBP Size	NSN
X-Small	TBD
Small-Short	TBD
Small	TBD
Medium	TBD
Large	TBD
X-Large	TBD

APPENDIX A
IMBP Testing Protocol

<u>V_{50} Ballistic Limit Measurements</u>

V_{50} ballistic limit testing of IMBPs shall be conducted as per paragraph 4.8.6 of this PD and MIL-STD-662. The IMBP will yield the minimum V_{50} ballistic limit as outlined in Table A.1 below. Threats and supporting information are outlined in Appendix F.

Table A.1: V_{50} Ballistic Limit Protocol

Threat	**Shots Per Plate**
RED	3
ORANGE	3
YELLOW	3
GREEN	3
BLUE	3

Ballistic limit measurements will be performed on the actual IMBP. Shot spacing for any part of impacted round should be no closer than 0.75-inches from an edge and each subsequent shot shall be spaced 5.0-6.0 inches from any previous shot.

V_{50} Calculation: The arithmetic mean of three (3) Partial Penetration (PP's) and three (3) Complete Penetrations (CP's) within a 125 ft/sec velocity spread or four (4) Partial Penetration (PP's) and four (4) Complete Penetrations (CP's) within a 150 ft/sec velocity spread yield the minimum allowable V_{50} BL determination that will be accepted as reliable test results.

If 6 partial penetrations occur that are above the minimum V_{50}, the plate demonstrates a V_{50} above the minimum, the plate passes, and that V_{50} test is terminated. All shots' velocities are recorded.

<u>V_0 Ballistic Resistance</u>

Ballistic V_0 testing of IMBP shall be conducted on a recurring basis per paragraph 3.8.3 and 4.8.7. Appendix B outlines IMBP V_0 performance requirements.

The ballistic V_0 testing of IMBP shall have a minimum of 3 impacts (2 impacts at 0-degree obliquity and 1 impact at 30-degree obliquity). The first two shots must be fired at 0-degree obliquity and the third shot at 30-degree obliquity. For the 30-degree obliquity shots, the direction of the obliquity depends on which side of the centerline is the intended impact point. If on the right, that side shall be rotated up-range (toward the gun barrel); if left, that side shall be rotated up-range. One shot shall be an edge shot (0-degree obliquity) between 0.75 and 1.25 inches from any edge (see Figure A.1); one shot shall be a crown shot (0-degree obliquity) 5.0-6.0 inches from any other shot; the third shot (30-degree obliquity) 5.0-6.0 inches from any other shot. Back face deformation measurements, per paragraph 4.8.9, will be taken on the first two shots, and UTL computed. Back face deformation measurements, per paragraph 4.8.9, will be taken on the last shot for government reference.

Blunting/Shatter Gap Resistance

Appendix B outlines IMBP blunting gap performance requirements.

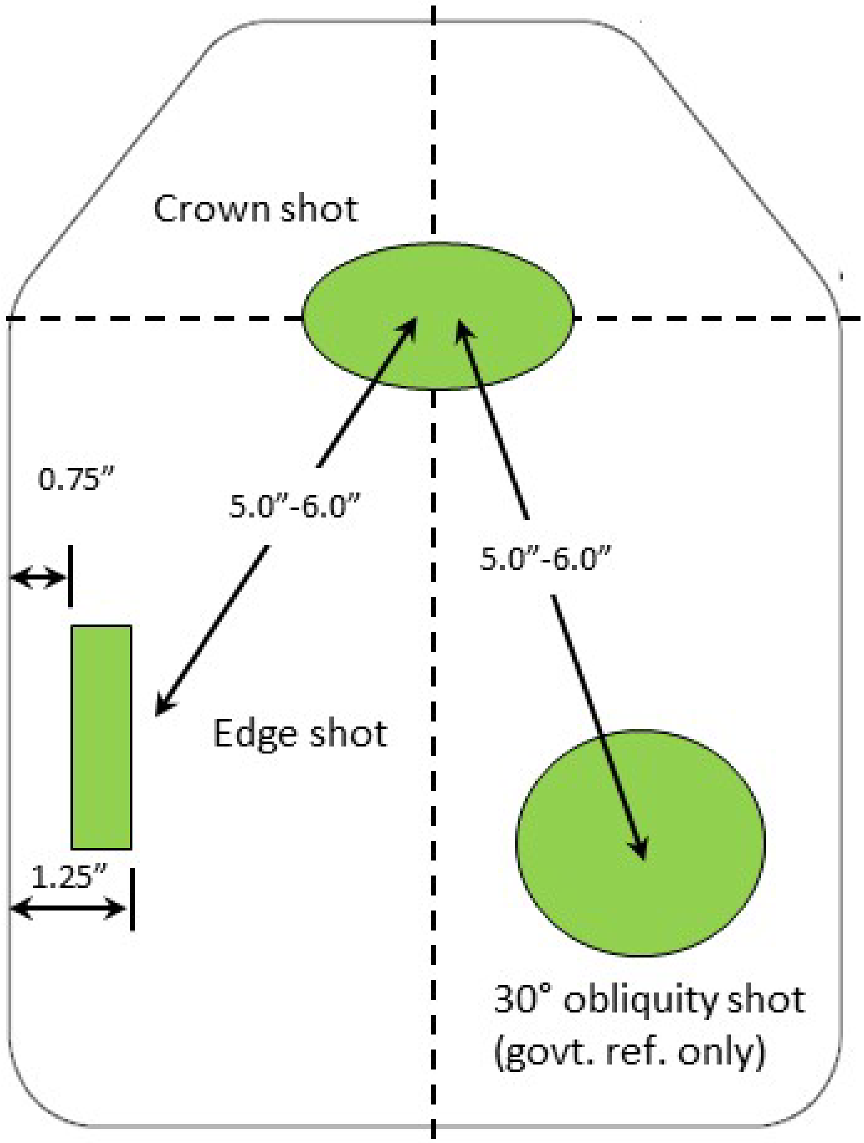

FIGURE A.1: General Shot Pattern (not to scale).

APPENDIX B
IMBP First Article Test (FAT) Protocol

B.1. First Article Testing. This Appendix is a mandatory part of the specification and provides the detailed protocol for the execution of First Article Testing (FAT). FAT ensures that the contractor may furnish a ballistic plate that conforms to all contract requirements for acceptance. FAT samples shall be manufactured using the same materials, equipment and procedures used during production and the FAT samples shall serve as the manufacturing standard. FAT may be waived by the Government when supplies identical or similar to those called for have previously been tested and/or delivered by the offeror or contractor and were accepted by the Government.

FAT for the IMBP shall be conducted in two (2) parts. Part 1 shall be in accordance with the DOT&E protocol and Part 2 shall consist of additional testing required to meet service-specific requirements. First Article Testing shall be conducted at Aberdeen Test Center (ATC). In order to gain FAT approval, a submitted design shall successfully pass the requirements outlined in FAT Part 1 and Part 2. V0 and blunting/shatter gap testing will be conducted with NIJ Type IIIA/HG2 shootpacks. Failure to meet the requirements of any sub-test or part thereof shall result in rejection of the First Article Test. Table B.1 outlines IMBP FAT protocol. Table B.2 outlines shot order and location for V0 and shatter gap testing. Threats and supporting information are outlined in Appendix F.

Note: Shot tolerance disputes shall be addressed by the vendor's representative at the time of shot in question. The government will not accept shot disputes after test results have already been released.

B.2 Requirements – FAT Part 1. Table B.3 represents the resistance to penetration and BFD statistical analysis required for both torso and side plates for FAT Part 1 testing. For resistance to penetration, the LCL for the P(nP) is the statistic of interest and the result compared against a 90% P(nP) for first shot (Torso and Side Plates) and 70% P(nP) for the second shot (Torso Plates Only). For BFD, the UTL shall be computed using BFD as a continuous normal random variable and the result compared against the requirement.

B.2.1 Resistance to Penetration. For each valid test shot, the result is determined to either experience a system complete penetration or a system partial penetration as defined in section 6. The numbers of system complete and system partial penetrations shall be used to calculate the P(nP) The Lower Confidence Level (LCL) is calculated for the first and second shots by combining shot locations, plate sizes and environmental conditions. The LCL must meet or exceed the requirement specified in Table B.3.

B.2.2 Back Face Deformation Analysis. For BFD, the metric of merit is a one sided UTL based on the assumption of normally distributed BFD data; the UTL to be calculated as specified in Table B.3 The UTLs are calculated for the first and second shots by combining shot locations, plate sizes and environmental conditions as identified in section 3.7.

B.2.3 Fair Hit/No Test Criteria. Fair hit/no test criteria are defined as outlined in Table B.4 below. In the case of an under-velocity shot which results in either a system complete penetration or a BFD greater than 44mm, the shot result shall be included in the analysis. If the under-velocity shot occurs on the first shot, the plate shall be replaced with a contingency plate to ensure a completed test matrix.

Table B.1: FAT Protocol

TEST	Threat	Plates	Spec Paragraph
PART 1 – DOT&E COMPLIANCE TESTING			
Ambient V_0	ORANGE	1 XS, 1 SS, 1 SM, 1 MD, 1 LG, 1 XL	3.8, 4.8, Appendix F
Temperature Shock V_0		1 XS, 1 SS, 1 SM, 1 MD, 1 LG, 1 XL	3.7, 3.8, 4.7, 4.8, Appendix F
JP-8 V_0		1 XS, 1 SS, 1 SM, 1 MD, 1 LG, 1 XL	
Oil V_0		1 XS, 1 SS, 1 SM, 1 MD, 1 LG, 1 XL	
Salt Water V_0		1 XS, 1 SS, 1 SM, 1 MD, 1 LG, 1 XL	
Weathered V_0		1 XS, 1 SS, 1 SM, 1 MD, 1 LG, 1 XL	
High Temp V_0		1 XS, 1 SS, 1 SM, 1 MD, 1 LG, 1 XL	
Low Temp V_0		1 XS, 1 SS, 1 SM, 1 MD, 1 LG, 1 XL	
Altitude V_0		1 XS, 1 SS, 1 SM, 1 MD, 1 LG, 1 XL	
Impacted V_0		1 XS, 1 SS, 1 SM, 1 MD, 1 LG, 1 XL	3.6, 4.6, Appendix F
PART 2 –V50 AND ADDITIONAL THREAT TESTING			
V_{50}, 0°	RED	2 XS, 2 MD, 2 LG	3.8, 4.8, Appendix F
	ORANGE	2 SM, 2 MD, 2 XL	
	YELLOW	2 SS, 2 SM, 2 XL	
	GREEN	2 XS, 2 SS, 2 LG	
Ambient V_0	YELLOW	3 XS, 3 SS, 3 SM, 3 MD, 3 LG, 3 XL	
Shatter Gap	ORANGE	16 MD	3.8, 4.8, Appendix F, Table B.6
Contingency		4 XS, 4 SS, 4 SM, 4 MD, 4 LG, 4 XL	
Buoyancy		4 XS, 4 SS, 4 SM, 4 MD, 4 LG, 4 XL	3.7.6, 4.7.6, Appendix F
Government Record		1XS, 1 SS, 1 SM, 1 MD, 1 LG, 1 XL	
Total		26 XS, 26 SS, 26 SM, 42 MD, 26 LG, 26 XL	

*Plates will sustain three shots. If all six shots exceed V50 minimum requirements, with no complete penetrations occurring, the requirement is considered met.

NOTES:

1. All shatter gap testing will be conducted with shoot packs.
2. Threats and supporting information are outlined in Appendix F.

Table B.2. FAT Protocol (V_0) – Shot Order and Location

Environment	**1st Shot Edge**	**1st Shot Crown**
PART 1 – DOT&E COMPLIANCE TESTING		
Ambient (Unconditioned)	XS, SS, LG	SM, MD, XL
Temperature Cycling	MD, LG, XL	XS, SS, SM
JP-8 Soak	XS, SS, SM	MD, LG, XL
Oil Soak	SM, MD, LG	XS, SS, XL
Salt Water	SS, MD, XL	XS, SM, LG
Weathered	SM, MD, LG	XS, SS, XL
High Temperature	XS, SS, SM	MD, LG, XL
Low Temperature	XS, SM, XL	SS, MD, LG
Altitude	XS, SS, MD	SM, LG, XL
Total	27	27
Impacted*	XS, SS, SM , MD, LG, XL	
Total	60	
PART 2 –ADDITIONAL THREAT TESTING		
Ambient (Unconditioned)	XS, SS, LG	SM, MD, XL
	MD, LG, XL	XS, SS, SM
	XS, SS, SM	MD, LG, XL
Total	18	

NOTE: All V0 testing will be conducted with shoot packs.

*For Impact-Conditioned plates, two shots per plate will be taken: first shot is at the location of the most severe crack, as determined by x-ray. If a crack is not visible after x-ray of the plate, the 1st shot will be taken at the center of the plate (impact location). The second shot is at any edge.

Table B.3. RTP and BFD Requirements for FAT Part 1

Shot Number	**Resistance to Penetration (RTP) Requirement**	**Backface Deformation (BFD) Requirement**
Shot 1	Lowerbound probability of P(nP) of .90 at a 90% confidence level	90% Upper Tolerance Limit on BFD with 90% confidence
Shot 2	Lowerbound probability of P(nP) of .70 at a 90% confidence level	80% Upper Tolerance Limit on BFD with 90% confidence

Table B.4. Fair Hit/No Test Criteria

Impact Velocity	Test Result		Evaluator Accepts or Rejects for Inclusion in Analysis Resistance to Penetration (RTP) Requirement		Proceed to next data point for that plate?
	Penetration	BFD	Penetration	BFD	
Acceptable	No Penetration (PP and CP)	Measured	Include as Success	Include	Yes
Acceptable	Complete System Penetration (CC)	Not Measured	Include as Failure	Not Measured	Yes
Too High	No Penetration (PP and CP)	Measured	Not Included	Not Included	No
Too High	Complete System Penetration (CC)	Not Measured	Not Included	Not Included	No
Too Low	No Penetration (PP and CP)	Measured	Not Included	Not Included is ≤ 44.0mm Included if > 44.0 mm	No
Too Low	Complete System Penetration (CC)	Not Measured	Include as Failure	Not Measured	No

B.2.4. FAT Part 1 Rejection Criteria. FATs that fail either the resistance to penetration or the BFD criteria shall be rejected. A submitted FAT shall be rejected if one or more of the following criteria are met:

- Calculated P(nP) less than 90% on first shot.
- Calculated P(nP) less than 70% on second shot.
- 90% UTL with 90% confidence > 44.0 mm on first shot
- 80% UTL with 90% confidence > 44.0 mm on second shot

B.3. Requirements – FAT Part 2. FAT Part 2 shall be conducted in accordance with the protocol outlined in table B.1 to verify the requirement outlined in paragraph 3.8.

B.3.1. V0 Requirements. Table B.5 represents the resistance to penetration and BFD statistical analysis required for FAT Part 2 testing. For resistance to penetration, the LCL for the P(nP) is the statistic of interest and the result compared against a 90% P(nP) for first shot and 90% P(nP) for the second shot. Only system complete penetrations are considered a failure. For BFD, no measured BFD shall be greater than 44.0 mm. Resistance to penetration shall be measured and analyzed in accordance with paragraph 3.8.3. BFD shall be measured in accordance with 4.8.9. Fair hit criteria is defined in Table B.4 above.

Table B.5. RTP and BFD Requirements for FAT Part 2

Shot Number	**Resistance to Penetration (RTP) Requirement**	**Backface Deformation (BFD) Requirement**
Shot 1	Lowerbound probability of P(nP) of .90 at a 80% confidence level	No BFD > 44.0 mm
Shot 2	Lowerbound probability of P(nP) of .90 at a 80% confidence level	No BFD > 44.0 mm

B.3.2 V50 Requirements. The determined V50 shall meet or exceed the minimum V50 as required in paragraph Table B.1 for the specified threats when tested in accordance with Appendix A of this specification.

B.3.3 Shatter Gap Requirements. The protocol and requirements for shatter gap are provided in Table B.6. Shot pattern shall alternate between edge and crown.

B.3.4 FAT Part 2 Rejection Criteria. Tests that fail either the resistance to penetration or the BFD criteria shall be rejected. A submitted FAT shall be rejected if one or more of the following criteria are met:

- Calculated P(nP) less than 90% on first shot for V0 tests.
- Calculated P(nP) less than 90% on second shot for V0 tests.
- BFD > 44 mm for V0 tests.
- Any V50 less than the minimum required for V50 tests.
- Any system complete penetration for shatter gap tests.
- Any BFD > 44 mm for shatter gap tests.

Table B.6: Shatter Gap Requirements

Threat	Velocity Range (ft./s)	Number Shots	Requirement
ORANGE	1800-1900	1 at 0°obliquity	Test 4 plates and obtain 4 first shot partial penetrations and acceptable BFD (≤ 44mm). Any system complete penetrations or unacceptable BFD is a fail and additional plates shall be tested.
	1901-2000	1 at 0°obliquity	Test 4 plates and obtain 4 first shot partial penetrations and acceptable BFD (≤ 44mm). Any system complete penetrations or unacceptable BFD is a fail and additional plates shall be tested.
	2001-2100	1 at 0°obliquity	Test 4 plates and obtain 4 first shot partial penetrations and acceptable BFD (≤ 44mm). Any system complete penetrations or unacceptable BFD is a fail and additional plates shall be tested.
	2101-2200	1 at 0°obliquity	Test 4 plates and obtain 4 first shot partial penetrations and acceptable BFD (≤ 44mm). Any system complete penetrations or unacceptable BFD is a fail and additional plates shall be tested.

NOTES:

1. All shatter gap testing will be conducted with shoot packs.
2. Threats and supporting information are outlined in Appendix F.

Appendix C
IMBP Lot Acceptance Test (LAT) Protocol

C.1 Lot Acceptance Testing. This Appendix is a mandatory part of the specification and provides the detailed protocol for the execution of Lot Acceptance Testing (LAT). Unless waived by the Government, satisfactory LAT is required prior to the delivery of the finished ballistic plate lot to the destination on the contract. Lot acceptance testing shall be conducted on a random sample of finished ballistic plates and any failure to meet the requirements of any test or sub-test shall result in rejection of the lot.

All test samples shall be randomly selected from the finished production lot by a Government representative (Contract Officer, Contracting Officers Representative, DCMA, or other representative as specified in the contract). The ballistic lot test data shall NOT be considered valid or accepted by the Government if the samples were not selected by a Government representative and a DD1222 was not signed.

If a lot fails the required testing, that lot is rejected in total and no component parts may be used in the production of any other lot or shipped to the destination on the contract. All failed lot acceptance tests shall require a failure analysis and corrective action report (FACAR). If the contractor chooses to submit a new design as a result of the failure analysis a First Article Test in accordance with Appendix A shall be required at the expense of the contractor. The Government reserves the right to Terminate for Default the contract in lieu of acceptance of the FACAR.

Note: Shot tolerance disputes shall be addressed by the vendor's representative at the time of shot in question. The government will not accept shot disputes after test results have already been released.

C.2 Protocol. LAT shall be conducted in accordance with Table C.1 for Threat ORANGE with a NTV shootpack. Test set-up shall be in accordance TOP 10-2-210. Test quantities and sample sizes are based on ANSI/ASQ Z1.4-2008 Special Inspection Level "S-4." Switching rules do not apply unless authorized by the Government. Buoyancy samples are per size plate produced for any given Lot.

Table C.1. Lot Sample Size

Lot Size	**ANSI/ASQ Z1.4 Code Letter**	**Sample Size**	**Buoyancy**	**Contingencies**	**Total Plates**
<150	D	8	3	3	14
151-500	E	13	3	3	19
501-1200	F	20	3	4	27
1200-3200	G	32	3	6	41

C.2.1 Test Set Up and Shot Spacing. All tests shall be conducted in accordance with section 4.8.7 of this specification. All LATs shall be 3-shot, V0 Testing of each insert against threat ORANGE– 3rd shot (30 degree obliquity) is for government reference only. Shot locations per Figure A.1. The shot pattern shall alternate between first shot edge, second shot crown, and first shot crown, second shot edge until all sample plates are tested.

C.3 Requirements. Table C.2 shows the resistance to penetration and backface deformation criteria required for the IMBP LAT protocol.

Table C.2. RTP and BFD Requirements for LAT

1st Shot	2nd Shot
Resistance to Penetration	
No 1st shot Complete System Penetration allowed for any lot size	10% Acceptance Quality Limit
Back Face Deformation [≤ 44.0 mm]	
80% Upper Tolerance Limit with 90% Confidence	70% Upper Tolerance Limit with 90% Confidence

For resistance to penetration, no first shot penetrations of are allowed. The second shot Acceptance Quality Limit is 10%. The overall resistance to penetration requirement is shown in Table C.3.

Table C.3. RTP Requirement Accept/Reject Criteria

Lot Size	ANSI/ASQ Z1.4 Code Letter	Sample Size	1st Shot System Complete Penetration		2nd Shot System Complete Penetration	
			Accept	Reject	Accept	Reject
91-150	D	8	0	1	2	3
151-500	E	13	0	1	3	4
501-1200	F	20	0	1	5	6
1201-3200	G	32	0	1	7	8

C.4 Lot Rejection Criteria. A submitted IMBP lot shall be rejected if one or more of the following criteria are met:

- One or more system complete penetrations on first shot.
- For second shot, an overall system complete penetration rate higher than allowed by Table C.2 (derived ANSI/ASQ Z-1.4-2008 using Acceptance Quality Limit (AQL) of 10)
- 80% UTL with 90% Confidence > 44.0 mm on first shot
- 70% UTL with 90% Confidence > 44.0 mm on second shot

Appendix D
IMBP Workmanship and Dimensional Tolerance Grading for Acceptance

The following dimensional criteria apply to all dimensional measurements except thickness. Thickness criteria is outlined in paragraph 3.5.4 and any measurements that exceed maximum thickness or tolerance for variability in thickness shall be recorded as a critical defect. Workmanship defect classification is outlined in Table D.1.

Workmanship and dimensional defects shall be assessed separately from ballistic testing defects for the purposes of acceptance determination.

Accept/reject criteria apply uniquely to each plate size.

Table D.1. Classification of Defects

Defect	**Critical**	**Minor**
Ceramic Component		
Any Crack	X	
Pit or void greater than: 0.060 inch in depth or diameter	X	
Pit or void less than or equal to 0.060 inch in depth and diameter		X
Rivulet, pit or void over 2.5 inches in length.	X	
Rivulet , pit or void greater than 0.75 inch in length but no more than 2.5 inch with max width and max depth no greater than 0.060 inch		X
Edge chip greater than 1/2 inch in the face or width of ceramic	X	
Edge chip greater than 1/16 inch but not more than ½ inch in the face or width of ceramic		X
Edge chip greater than 1/8 inch in depth of ceramic	X	
Edge chip greater than 1/16 inch in depth of ceramic but not more than 1/8 inch		X
All Components		
Any delamination areas larger than 0.20 $inch^2$	X	
Any delamination areas less than or equal to 0.20 $inch^2$		X
Other		
Any foreign object debris or defect identified that is not associated with the Technical data package		X
Any Item larger than 0.04 inch in any dimension	X	
Any item equal to or less than 0.04 inch in any dimension		X

<u>Table D.1 Definitions:</u>

Cracks. Well-defined line or lines having sharp terminal points indicating a break in the ceramic material.
Void. Round or elongated, smooth edged spots occurring individually or randomly distributed within the ceramic indicating regions of missing material.

Pit. A region of missing material, or void, on the surface of the ceramic.
Rivulet. Rounded void with gradual, river like meandering, which make it distinguishable from cracks
Delamination. A region where there is no bond or material separation between single or multiple layers of the plate components, creating a gap in the materials.

Table D.1 Notes:
-Reject all plates with any critical defects or 3 or more minor defects.
-Any two minor defects less than 3/8 inch apart shall be classified as a critical defect.
-If any defect falls into two categories of defects, the more severe defect category will apply and only be considered one defect.
- Appendix D outlines overarching acceptance criteria for workmanship and dimensional defects.
-If applicable, this requirement should not supersede more stringent vendor internal requirements.

Critical dimensional defects:

- Any dimensional measurement that exceeds upper or lower tolerances by more than 0.02 inches.

Minor dimensional defects:

- Any dimensional measurement that exceeds upper or lower tolerances by 0.01 inches.
- Any dimensional measurement that exceeds upper or lower tolerances by 0.02 inches (weighted 1.5 times for each occurrence).

Accept / Reject criteria for FAT:

- Two or more critical defects constitutes FAT rejection.
- A minor defect rate of 14 or higher constitutes FAT rejection. Truncate after the decimal point of the cumulative minor defect rate.

Accept / Reject criteria for LAT:

- Two or more critical defects constitutes LAT rejection.
- A minor defect rate of 7 or higher constitutes LAT rejection. Truncate after the decimal point of the cumulative minor defect rate.

Appendix E
IMBP Buoyancy Verification

A. The buoyancy of the finished IMBP shall be measured as described below to determine compliance with the performance criteria outlined in paragraph 3.7.6.

1. This test is to be conducted in a test tank of water that is secured against change of water level or disturbance of the sample.

2. Air entrapped in folds of cloth or other areas is to be removed from the device immediately following submersion.

3. A test basket made of wire mesh or equivalent material and of sufficient size to hold the sample without unduly compressing the device is to be ballasted with sufficient weight to permit the complete submergence of the basket and device.

4. The ballasted basket is to be suspended from a scale calibrated to an accuracy of ±0.1 lbs., and the weight of the submerged basket is to be determined.

5. The IMBP is to be placed in the basket so that its upper surface is approximately 2 inches (51 mm) below the water's surface and is to remain submerged for a minimum for 24 hours.

6. The buoyancy of the IMBP is to be computed by subtracting the submerged weight of the ballasted basket and IMBP from the submerged weight of the ballasted basket alone.

7. The result is to be corrected as necessary to establish the buoyancy at an atmospheric pressure of 29.92 inches Hg (101 kPa) and a water temperature of 68°F (20°C).

B. Buoyancy Correction for Water Temperature and Atmospheric Pressure. The buoyancy shall be corrected using one of the following formulas.

1. Standard Formula with Atmospheric Pressure measured in millimeters of Mercury and Temperature measured in Degrees Celsius:

$$\textit{Buoyancy} \times \frac{\textit{Barometric Pressure (mmHg)}}{760.2\ (\textit{mmHg})} \times \frac{293.15\ °K}{\textit{Temp in } °C + 273.15\ °C}$$

2. Standard Formula with Atmospheric Pressure measured in Kilopascals and Temperature measured in Degrees Celsius:

$$\textit{Buoyancy} \times \frac{\textit{Barometric Pressure (kPa)}}{101.4\ kPa} \times \frac{293.15\ °K}{\textit{Temp in } °C + 273.15\ °C}$$

3. Standard Formula with Atmospheric Pressure measured in inches of Mercury and Temperature measured in Degrees Fahrenheit:

$$\textit{Buoyancy} \times \frac{\textit{Barometric Pressure (in. Hg)}}{29.92\ (\textit{in. mHg})} \times \frac{527.7\ °R}{\textit{Temp in } °F + 459.7\ °F}$$

4. Standard Formula with Atmospheric Pressure measured in PSI and Temperature measured in Degrees Fahrenheit:

$$\text{Buoyancy} \times \frac{\text{Barometric Pressure (PSI)}}{14.7\ \text{(PSI)}} \times \frac{527.7\ °\text{R}}{\text{Temp in } °\text{F} + 459.7\ °\text{F}}$$

The measured buoyancy of samples must be equal to or higher that zero (tolerance -0.1 lbs) for all IMBP sizes.

Appendix F
Threat Matrix

F. 1 Classified Information. Omitted from this file to enable overall file portability. Available to eligible parties under a separate cover and via authorized transmission or delivery methods.